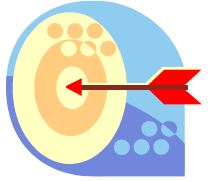


Lesson 12: Best Practices

Objectives



This lesson will introduce you to some best practices that you can adopt when working with Sterling B2B Integrator.

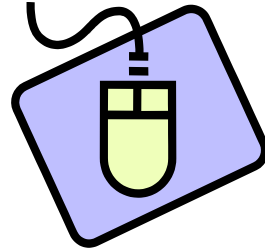
This lesson has been designed to enable you to:

- Implement best practices for Assign Service
- Implement best practices for Invoke Sub-Process Service
- Implement best practices for Release Service

Best Practices for Assign Service

Assign Service vs Assign Activity:

- The Assign Service can be used when there are multiple assignments that need to be made consecutively.
- Reduces the number of steps in a process thus increasing performance. A recent test using two assign statements vs. the same values in an Assign service ran in approximately half the time.
- The Assign Service can be found in the Sync Mode Stencil.



Exercise 13-1: Create an Assign Service Process

Your turn!

Pause this presentation, and follow the instructions on creating an Assign service process.

Best Practices for the Invoke Sub-Process Service

How the Invoke Sub-Process Service Works

- When the Invoke Sub-Process service is set to synchronous mode, the parent suspends processing until it receives data from the child. In synchronous mode, the parent is notified when the child encounters errors.
- When the Invoke Sub-Process service is set to asynchronous mode, the parent and child process data simultaneously and independently of each other. Therefore, the parent does not receive notification when the child encounters errors.
- When the Invoke Sub-Process service is set to run a sub-process inline, the Sub-process runs as part of the parent process, sharing the same process data.
- When the Invoke Sub-Process service is set to run in embedded mode, the Sub-process runs with no persistence, meaning that no record of the process is recorded in Application and no tracking is done.

Note: The **Invoke Sub-Process** and **Invoke Business Process Service** are different names for the same service used to start a child business process.

What Mode Should I Use?

- Asynchronous mode is the default selection and typically used in most applications as it allows for concurrent (multi-threaded) processing of multiple documents.
- Synchronous can be used when you want the child to report back up to the parent or if you want a single threaded process flow. An example of this may be large files such as Catalogs (832) or Product Activity Data (852) where due to the file size, handling multiple files at once can put an undue strain on the environment.
- Inline is a good choice when dealing with high volumes of processes within the system as it minimizes the number of threads that are created for the overall process flow.

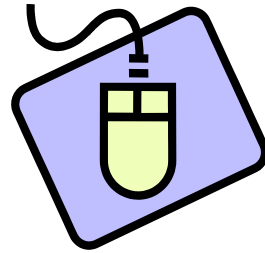
Performance Tips

By default, when you use the Invoke Sub-Process service in a business process, all process data passes from the parent process to its subprocess.

However, if you are using the Invoke Sub-Process service in sync mode, a special tag called ``message_to_child/message_to_parent'` allows you to pass along only the ``message_to_child/message_to_parent'` node in the process data of the parent process or subprocess. Using this tag can provide significant performance improvement.

Before invoking a subprocess, create a special tag called ``message_to_child'` in the parent process, and append all of the data needed in the subprocess under this tag. The Invoke Sub-Process service will pass only this tag to the subprocess.

The ``message_to_parent'` tag accomplishes the same task, but limits the amount of data that is returned to the parent process. Only data under the ``message_to_parent'` tag will be passed back to the parent.



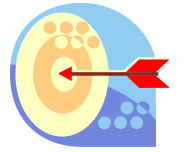
Exercise 13-2: Using the Invoke Sub-Process Service in Sync Mode Instructions

Your turn!

Pause this presentation, and follow the instructions on using the Invoke Sub-Process Service in sync mode.

Best Practices for Release Service.

Best Practices for Release Service

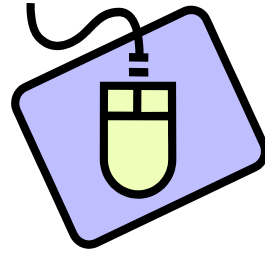


Overview:

The Release Service is a system service that is used to remove extraneous data from the process data.

This service comes with no predefined parameters, therefore you must use the Advanced Service Editor to create a parameter called TARGET and assign a value to it. The value identifies the process data which is no longer needed. TARGET is the name of the parameter the Release service expects in the "Message TO Service." The value assigned to the parameter is an XPath expression which selects the nodes from the Process Data to be released. Even though it is an XPath expression you do not need to check the "Use XPATH" checkbox.

When dealing with large amounts of Process data, the Release service can be used to clean up portions of Process Data that are no longer needed which aid in the performance of the business process. Multiple Process Data tags can be released within one Release Statement to reduce the overall number of steps in the BP.



Exercise 13-3: Re-execute the Invoke_Test Process
And

Exercise 13-4: Adding the Release Service to the
BasicInventoryProcess

Your turn!

Pause this presentation, and follow the instructions on re-executing the Invoke_Test process and Adding the Release Service to the Basicinventoryprocess.



Congratulations

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This lesson was designed to enable you to:

- Implement best practices for Assign Service
- Implement best practices for Invoke Sub-Process Service
- Implement best practices for Release Service